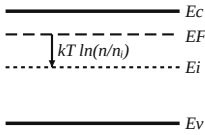
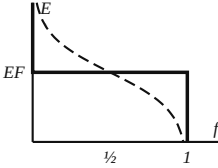


1 · Carriers at equilibrium

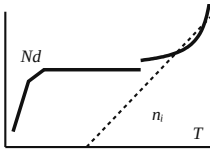
- $n_0 p_0 = n_i^2$ always at equilibrium.
- Majority = doping; minority = $n_i^2/\text{majority}$.
- $n_0 = n_i e^{(E_F - E_i)/kT}$, $p_0 = n_i e^{(E_i - E_F)/kT}$.
- $E_F - E_i = kT \ln \frac{n_0}{n_i}$ (ln). $\times 10$ in $n_0 \rightarrow E_F$ up 0.0596 eV.
- $n_0 = N_c e^{-(E_c - E_F)/kT}$, $p_0 = N_v e^{-(E_F - E_v)/kT}$. Si: $N_c = 2.8 \times 10^{19}$, $N_v = 1.04 \times 10^{19}$.
- $n_i = \sqrt{N_c N_v} e^{-E_g/2kT}$; $N_c \propto T^{3/2}$. Valid if E_F is $> 3kT$ from both band edges.
- Donors: col. V (P, As, Sb). Acceptors: col. III (B, Al, Ga). All ionized at 300 K.

**2 · Fermi function**

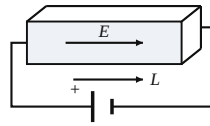
- $f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$: probability a state is full. Empty: $1 - f$.
- $f(E_F) = \frac{1}{2}$ at every T . $T = 0$: step.
- Symmetric: full at $E_F + \Delta$ = empty at $E_F - \Delta$.
- $E - E_F > 3kT$: $f \approx e^{-(E - E_F)/kT}$.

**3 · Compensation, neutrality, temperature**

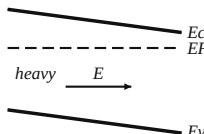
- $p_0 + N_d^+ = n_0 + N_a^- \rightarrow n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$.
- $|N_d - N_a| \gg n_i$: $n_0 = N_d - N_a$. **Else (Ge, hot Si, $N_d \approx N_a$) use the root.**
- $\frac{n_i(T_2)}{n_i(T_1)} = \left(\frac{T_2}{T_1}\right)^{3/2} e^{-\frac{E_g}{2k} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)}$, $k = 8.62 \times 10^{-5} \text{ eV/K}$.
- Freeze-out (cold, $n_0 < N_d$) $\rightarrow$ extrinsic ($n_0 = N_d$) $\rightarrow$ intrinsic (hot, $n_0 \rightarrow n_i$).

**4 · Drift, conductivity, resistance**

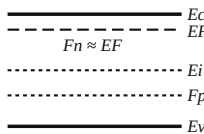
- $v_d = \mu \mathcal{E}$; $J = q(n\mu_n + p\mu_p)\mathcal{E} = \sigma \mathcal{E}$; $\rho = 1/\sigma$; $R = \rho L/A$.
- Both carriers' currents point along $\mathcal{E}$: they add.
- $\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$; $\mu_L \propto T^{-3/2}$; $\mu_I \propto T^{3/2}/(N_d + N_a)$.
- Light doping: μ falls with T . Heavy: rises then falls. More doping (even compensating): lower μ .
- High field: v_d saturates $\approx 10^7 \text{ cm/s}$. Hall: $p_0 = \frac{I_x B_z}{q t V_H}$.
- Si: $\mu_n = 1350$, $\mu_p = 480 \text{ cm}^2/\text{V}\cdot\text{s}$ (light doping).

**5 · Diffusion, total current, built-in field**

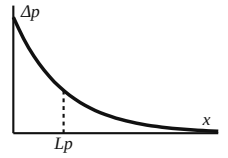
- $J_n = q\mu_n n \mathcal{E} + qD_n \frac{dn}{dx}$; $J_p = q\mu_p p \mathcal{E} - qD_p \frac{dp}{dx}$.
- Electrons diffuse down the gradient, so their current points **up** it.
- Einstein: $D = \frac{kT}{q} \mu$. Si: $D_n = 35$, $D_p = 12.4 \text{ cm}^2/\text{s}$.
- Equilibrium: E_F flat, $J_n = J_p = 0$. Graded: $\mathcal{E} = -\frac{kT}{q} \frac{1}{n} \frac{dn}{dx}$; $N_d = N_0 e^{-x/L} \rightarrow \mathcal{E} = +\frac{kT}{qL}$ (to light side); N_a : $-\frac{kT}{qL}$ (to heavy side).
- $\mathcal{E} = \frac{1}{q} \frac{dE_c}{dx}$ (field points uphill in E_c). Contact: $qV_0 = kT \ln \frac{N_a N_d}{n_i^2}$.

**6 · Excess carriers, quasi-Fermi levels**

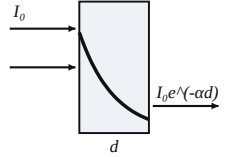
- Low level ($\delta n \ll n_0 + p_0$): $\delta n = \Delta n e^{-t/\tau}$, $\tau_n = \frac{1}{\alpha_r(n_0 + p_0)} \approx \frac{1}{\alpha_r p_0}$ (p-type).
- High level: $\delta n = \frac{\Delta n}{1 + \alpha_r \Delta n \tau}$, halves at $t = \frac{1}{\alpha_r \Delta n}$.
- Steady light: $\delta n = g_{op} \tau_n$, $\delta p = g_{op} \tau_p$. $g(T) = \alpha_r n_i^2$.
- $n = n_i e^{(F_n - E_i)/kT}$, $p = n_i e^{(E_i - F_p)/kT}$. Use $n_0 + \delta n$! Minority QFL moves.
- $\Delta \sigma = q g_{op} (\tau_n \mu_n + \tau_p \mu_p)$. $n p = n_i^2$ is **equilibrium only**.

**7 · Continuity, diffusion length**

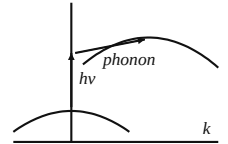
- $\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$; $\frac{\partial n}{\partial t} = +\frac{1}{q} \frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}$.
- No field: $\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$. Steady: $L_p = \sqrt{D_p \tau_p}$.
- Long bar: $\delta p = \Delta p e^{-x/L_p}$, $J_p = \frac{q D_p}{L_p} \Delta p e^{-x/L_p}$.
- Short bar ($W \ll L$): linear, $J_p = q D_p \Delta p / W$.

**8 · Optical absorption**

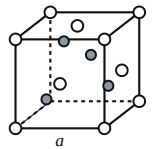
- $E [\text{eV}] = 1.24/\lambda [\mu\text{m}]$. Absorbed only if $h\nu \geq E_g$.
- $I = I_0 e^{-\alpha x}$; absorbed $1 - e^{-\alpha d}$. Heat fraction $\frac{h\nu - E_g}{h\nu}$. Photons/s = $P/h\nu$ (J).

**9 · Bands, effective mass**

- $m^* = \hbar^2 / \frac{d^2 E}{dk^2}$; $E = E_c + A k^2 \rightarrow m^* = \frac{\hbar^2}{2A}$ (J, m).
- Direct (GaAs): emits light. Indirect (Si, Ge): needs phonon.
- Metal: partly filled band. Semi: $E_g \sim 1 \text{ eV}$. Insulator: $\gtrsim 5 \text{ eV}$.
- Gaps: Si 1.12, GaAs 1.43, Ge 0.67 eV.

**10 · Crystals**

- Atoms/cell: sc 1, bcc 2, fcc 4, diamond 8. $N = \# / a^3$. Si $a = 5.43 \text{ \AA} \rightarrow 5.00 \times 10^{22}$.
- Miller: intercepts $\rightarrow$ reciprocals $\rightarrow$ integers. $[hkl] \perp (hkl)$ (cubic). (100) fcc/diamond: $2/a^2$.

**11 · Traps**

- $\mu\text{m} \rightarrow \text{cm}$: $\times 10^{-4}$. $\text{\AA} \rightarrow \text{cm}$: $\times 10^{-8}$. Use **ln**, not **log**.
- Minority is n_i^2/N , not n_i . Lifetime uses the **majority concentration**.
- $L = \sqrt{D\tau}$. Sign of J_n^{diff} : $+q D_n dn/dx$.

Ex 1 · Doped Si: n_0, p_0, E_F Si, $N_d = 10^{16}$, 300

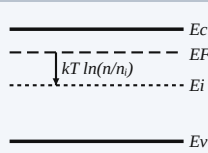
K.

$$n_0 = 10^{16}; p_0 =$$

$$\frac{2.25 \times 10^{20}}{10^{16}} = 2.25 \times 10^4 \text{ cm}^{-3}.$$

$$E_F - E_i = 0.0259 \ln \frac{10^{16}}{1.5 \times 10^{10}} = 0.347 \text{ eV}.$$

$$E_c - E_F = 0.0259 \ln \frac{2.8 \times 10^{19}}{10^{16}} = 0.206 \text{ eV}.$$

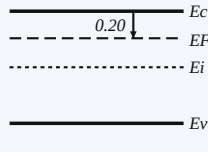
**Ex 2 · Fermi probability, n_0 from N_c** Si, E_F 0.20 eV below E_c .

$$f(E_c + kT) =$$

$$\frac{1}{1 + e^{0.2259/0.0259}} = 1.63 \times 10^{-4}.$$

$$n_0 = 2.8 \times 10^{19} e^{-7.72} = 1.24 \times 10^{16}.$$

$$p_0 = 1.04 \times 10^{19} e^{-0.92/0.0259} = 3.89 \times 10^3.$$

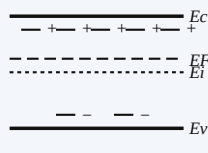
**Ex 3 · Compensated Ge (n_i not negligible)** $N_d = 3 \times 10^{13}$, $N_a = 10^{13}$, $n_i =$ 2.5×10^{13} .

$$n_0 = 10^{13} +$$

$$\sqrt{10^{26} + 6.25 \times 10^{26}} = 3.69 \times 10^{13}.$$

$$p_0 = n_i^2/n_0 = 1.69 \times 10^{13}. \text{ Check: } p_0 + N_d = n_0 + N_a \checkmark.$$

$$\text{Shortcut } N_d - N_a \text{ would be 46\% low.}$$

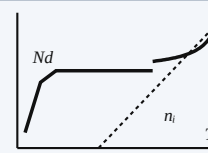
**Ex 4 · Hot Si** $n_i(600) = 1.5 \times 10^{10}$.

$$2^{3/2} e^{\frac{1.12}{2(8.62 \times 10^{-5})}} \cdot \frac{1}{600} = 2.14 \times 10^{15}.$$

$$N_d = 10^{15}; n_0 =$$

$$5 \times 10^{14} +$$

$$\sqrt{2.5 \times 10^{29} + 4.57 \times 10^{30}} = 2.70 \times 10^{15}.$$

**Ex 5 · Bar resistance**n-Si 10^{16} , $\mu_n =$ 1350, $L = 1 \text{ mm}$, $A = 100 \times 10 \mu\text{m}^2$,

5 V.

$$\sigma = qN_d\mu_n =$$

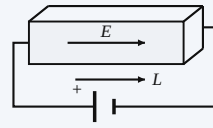
$$2.16 (\Omega\text{cm})^{-1}, \rho =$$

$$0.463 \Omega\text{cm}.$$

$$A = 10^{-5} \text{ cm}^2; R = \frac{0.463(0.1)}{10^{-5}} = 4.63 \text{ k}\Omega, I = 1.08 \text{ mA}.$$

$$J = 108 \text{ A/cm}^2, v_d = 1350(50) =$$

$$6.75 \times 10^4 \text{ cm/s}.$$

**Ex 6 · Mobility at another T** 300 K: $\mu_L = 1500$, $\mu_I = 3000 \rightarrow \mu =$

1000.

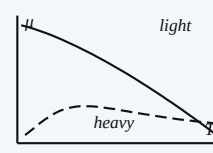
400 K: $\mu_L =$

$$1500(4/3)^{-3/2} =$$

$$974, \mu_I =$$

$$3000(4/3)^{3/2} = 4620.$$

$$\mu = (1/974 + 1/4620)^{-1} = 805 \text{ cm}^2/\text{V}\cdot\text{s}.$$

**Ex 7 · Two regions in contact** $N_a = 10^{16}$ meets $N_d = 10^{17}$.•n: $E_F - E_i =$

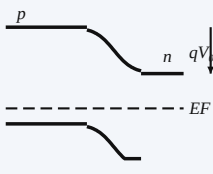
$$0.0259 \ln(6.67 \times$$

$$10^6) = 0.407; \text{ p:}$$

$$E_i - E_F = 0.347$$

$$\text{eV}.$$

$$E_F \text{ lines up; bands step } qV_0 = 0.754 \text{ eV}.$$

**Ex 8 · Graded doping** $N_d = 10^{17} e^{-x/L}$, $L = 0.5 \mu\text{m}$, $\mu_n =$

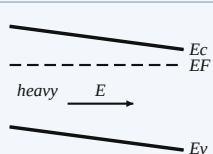
1000.

$$\mathcal{E} = \frac{kT}{qL} =$$

$$\frac{0.0259}{0.5 \times 10^{-4}} = 518$$

$$\text{V/cm, } +x.$$

$$D_n = 25.9. \text{ At 0: drift } +8.29 \times 10^3, \text{ diffusion } -8.29 \times 10^3 \text{ A/cm}^2; \text{ sum 0.}$$

**Ex 9 · Light into thin GaAs**0.5 μm , $\alpha = 3 \times$ 10^4 , 10 mW at 2.00eV, $E_g = 1.43$.

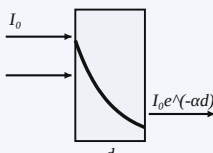
$$\alpha d = 1.5; P_{\text{abs}} =$$

$$10(1 - e^{-1.5}) =$$

$$7.77 \text{ mW}.$$

$$\text{Heat} = 7.77 \cdot \frac{0.57}{2.00} = 2.21 \text{ mW}.$$

$$\text{Photons } \frac{7.77 \times 10^{-3}}{3.2 \times 10^{-19}} = 2.43 \times 10^{16}/\text{s}.$$

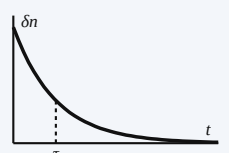
**Ex 10 · Pulse decay, low level**p-GaAs 10^{17} , $\alpha_r =$ 7.2×10^{-10} , $\Delta n =$ 10^{14} .

$$\tau_n = \frac{1}{\alpha_r p_0} = 13.9$$

ns. At 20 ns:

$$10^{14} e^{-20/13.9} = 2.37 \times 10^{13}.$$

$$\text{To } 10^{12}: t = 13.9 \ln 100 = 64.0 \text{ ns}.$$

**Ex 11 · High level** $n_0 + p_0 \approx 0$, $\alpha_r = 10^{-10}$, $\Delta n = 10^{17}$.

$$\delta n(100 \text{ ns}) = \frac{10^{17}}{1 + (10^{-10})(10^{17})(10^{-7})} = 5.00 \times 10^{16}.$$

Ex 12 · Quasi-Fermi levels, photoconductivityn-Si 10^{15} , $g_{op} =$ 10^{19} , $\tau = 2 \mu\text{s}$.

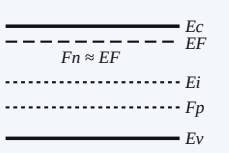
$$\delta p = g_{op} \tau =$$

$$2.00 \times 10^{13} \text{ (low}$$

level).

$$F_n - E_i = 0.0259 \ln(6.80 \times 10^4) = 0.288; E_i - F_p = 0.0259 \ln 1333 = 0.186 \text{ eV}.$$

$$\Delta \sigma = q \delta p (\mu_n + \mu_p) = 5.86 \times 10^{-3} (\Omega\text{cm})^{-1}.$$

**Ex 13 · Steady-state injection**

Holes into n-Si:

 $\Delta p = 10^{14}$, $\mu_p = 480$, $\tau_p = 1 \mu\text{s}$.

$$D_p = 12.4, L_p =$$

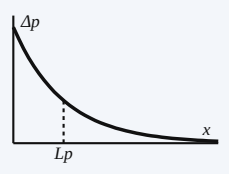
$$\sqrt{12.4 \times 10^{-6}} =$$

$$35.3 \mu\text{m}.$$

$$\delta p(50 \mu\text{m}) = 10^{14} e^{-50/35.3} = 2.42 \times 10^{13}$$

.

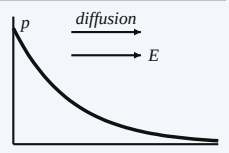
$$J_p(0) = \frac{q D_p \Delta p}{L_p} = 56.4 \text{ mA/cm}^2, \text{ toward } +x.$$

**Ex 14 · Drift + diffusion components** $p = 10^{16} e^{-x/2 \mu\text{m}}$, $\mathcal{E} = +20 \text{ V/cm}$, $\mu_p = 480$; at $x = 0$:

$$\text{Drift } q \mu_p p \mathcal{E} = 15.4$$

$$\text{A/cm}^2.$$

$$\text{Diffusion } -q D_p \frac{dp}{dx} = + \frac{q D_p p}{L} = 99.5 \text{ A/cm}^2. \text{ Total } 114.8 \rightarrow 115 \text{ A/cm}^2.$$

**Ex 15 · Atoms per cm^3 and per cm^2**

$$\text{Si: } 8/(5.43 \times$$

$$10^{-8})^3 = 5.00 \times$$

$$10^{22} \text{ cm}^{-3}; \rho_m =$$

$$\frac{NM}{N_A} = 2.33 \text{ g/cm}^3.$$

$$\text{(100): } 2/a^2 =$$

$$6.78 \times 10^{14} \text{ cm}^{-2}.$$

